

Vinay Kumar

Senior AI Engineer

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Professional Summary

Senior AI Engineer with 5.5+ years of experience in enterprise AI platform development, ML engineering, and Data Science. Designed and built foundational agent infrastructure, composed it into enterprise-grade products, and architected diverse AI solutions across multiple domains and clients at Infosys.

Professional Experience

Infosys — Senior AI Engineer

Dec 2024 – Present

Foundational Components

- **Execution Harness:** Designed and built a bespoke event-driven agent execution engine supporting deterministic state reconstruction, checkpointing, and long-running workflow continuity across distributed agent instances.
- **Sandbox:** Built an isolated execution sandbox leveraging gVisor for secure, reproducible execution of untrusted agent code, MCP servers, and dynamically loaded skills.
- **MCP Platform:** Created a platform for importing, building, and hosting Model Context Protocol (MCP) servers with AI-assisted authoring, validation, and lifecycle management.

Products

- **Process-Centric Harness:** Composed foundational components into a full product enabling end-to-end process authoring, skill linking, OpenAPI spec onboarding, and autonomous execution. Deployed across multiple enterprise clients and under active go-to-market.
- **User-Centric Harness:** Built a multi-surface developer platform — including Web Portal (AGUI/A2UI), CLI, and VS Code Extension — exposing agent capabilities for developer and operational workflows.

Solution Architecture & Deal Wins

- Independently architected diverse AI solutions — including a \$300M Telecom transformation, Digital Tutor platform, and CMDDB platform — driving multiple deal wins for Infosys.

Technical

- **LLM Compiler:** Built a proprietary LLM execution optimizer with dynamic prompt planning and semantic caching, reducing median query latency by 35%.

Xcaliber Infotech (A Phoenix Group Company) — Data Scientist / AI-ML Engineer

Dec 2022 – Dec 2024, Pune

- **Summarization Microservice:** Built a high-throughput text summarization service using LLaMA, BART, spaCy, and PyTorch to process OEM delivery waiver comments, integrating with warehouse ERP systems.
- **Enterprise RAG Assistant:** Developed a conversational assistant using LangChain, ChromaDB, and Hugging Face Transformers, reducing internal document lookup time by 40%.
- **Accessory Recommender:** Designed a location-aware affinity recommendation engine matching regional inventory with purchase patterns, boosting weekly accessory bundle sales by 25%.
- **API & Telemetry Layer:** Developed REST APIs and real-time operations dashboards for role-based analytics and audit tracking.

Concentrix India — Data Analyst

Oct 2015 – Apr 2017, Gurgaon

- **Analytics Pipelines:** Built automated Python and SQL ETL pipelines and executive reporting dashboards.
- **Predictive Modeling:** Performed regression and classification modeling for business forecasting and workforce planning.

Skills

- **AI Platform & Agent Systems:** Event-Driven Architecture, Agent Execution Harnesses, gVisor Sandboxing, Model Context Protocol (MCP), Process Orchestration, Multi-Agent Systems, Semantic Caching, LLM Compilers.
- **GenAI & LLM Engineering:** LLaMA, Mistral, GPT-4/Azure OpenAI, Hugging Face Transformers, BERT, BART, RoBERTa, LangChain, RAG Architectures, Vector DBs (ChromaDB, Pinecone, FAISS).
- **Machine Learning & Data Science:** PyTorch, TensorFlow, Scikit-Learn, spaCy, NLTK, NER, Regression, Classification, Clustering, Time-Series, Recommendation Systems.
- **Languages & Infrastructure:** Python, TypeScript, SQL (PostgreSQL, MySQL), FastAPI, Flask, Docker, Kubernetes, AWS (EC2, S3, SageMaker, Lambda), Git, CI/CD, Linux.

Education

PG-Diploma in Artificial Intelligence (PGDAI) — C-DAC, Pune, 2022

B.Tech in Mechanical Engineering — JIET, Jind, 2014